

	hence the rate the motor provides energy is $(m_1 + m_2)g\dot{y}$.
8.	Ans: B Horizontal component of tension provides centripetal force, so $T \sin 30^\circ = mr\omega^2, \text{ where } \omega = \frac{2p}{2} = p, \text{ and } r = 10 \sin 30^\circ$ $T \sin 30^\circ = 0.5 \cdot 10 \sin 30^\circ \cdot p^2$ $T = 49 \text{ N}$
9.	Ans: D